Jose Edgar Hernandez

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Deep learning engineer with four years training and deploying neural networks, an **MSc in Artificial Intelligence** (Erasmus Mundus EMAI) and a **BSc in Robotics**. I work on generative models, **vision-language-action (VLA)** and **diffusion**, from architecture design and editing to training and inference. I have put learned policies onto real hardware (UR5, xArm6, SO-ARM101, Jetson). First-author paper at IEEE IJCB 2026.

Education

2024 – 2026 **MSc in Artificial Intelligence (Erasmus Mundus EMAI)**
Pompeu Fabra University – Sapienza University – University of Ljubljana

2021 – 2024 **BSc in Robotics and Digital Systems Engineering**
Tecnológico de Monterrey GPA 96 /100

Research & Publications

2026 **Explicit Perceptual Guidance in Vision-Language-Action Models**, MSc Thesis
Visual Cognitive Systems Laboratory (ViCoS), University of Ljubljana View
Trained a visual-thought decoder over a frozen **X-VLA** policy to recover expert perceptual structure from its VLM representations (DINOv2 tokens, dense CeDiRNet grasp maps), used to condition the policy to improve manipulation. Full **LeRobot** loop: SO-ARM101 teleoperation, DDP fine-tuning, real-robot evaluation.

2026 **PIU: Proximity-guided Identity Unlearning in ID-Conditioned Diffusion Models**
J. E. Hernandez et al. *IEEE Int. Joint Conference on Biometrics (IJCB), 2026. First author.* View

Experience

Jan. – Feb. 2024 **Visiting Researcher – Kavraki Lab**
Rice University | Robotics and motion planning group, Houston, USA View
Integrated robot motion planning with Samsung's SceneGrasp model for 3D reconstruction and grasp pose estimation on a UR5 manipulator.

Mar. 2022 – Jan. 2026 **Machine Learning Engineer (Part-Time)**
Saveware | Remote. Attention-based sequence models with uncertainty estimation, and the data pipelines feeding them. View

Selected Robotics & ML Projects

2026 **Smart Pouring: Robot Imitation Learning from Phone Videos**
Humanoid robot learning challenge View
Fine-tuned **SmolVLA** (LeRobot, LIBERO) from 48 phone videos of my own pours, retargeted to robot demos via ArUco pose tracking. DART/HG-DAgger recovery data and a grasp-offset state memory took full-task success from 0% to 50% (96% grasp).

2026 **Visual-Thought Token Grounding in Frozen VLM Representations**
University of Ljubljana View
Scored visual-thought queries against the frozen X-VLA image-token grid for instruction-conditioned spatial relevance maps, refined coarse-to-fine over DaViT features.

2024 **Autonomous Localization and Navigation for a Differential Drive Robot**
Tecnológico de Monterrey View
Vision and LiDAR navigation stack with EKF odometry fusion and visual servoing, real-time on NVIDIA Jetson.

2023 **Deep Reinforcement Learning for Trajectory Planning of a Robotic Arm**
Tecnológico de Monterrey View
DDPG with Hindsight Experience Replay for goal-conditioned manipulation, in MuJoCo and on a real xArm6.

Technical Skills

Deep Learning: PyTorch, Hugging Face, Diffusers, CUDA
Training: LoRA/PEFT, distributed training (DDP), W&B
Robotics: LeRobot, ROS, MuJoCo, SLAM, NVIDIA Jetson
Languages: Spanish (native), English (C2), Italian (A2)
Models: X-VLA, SmolVLA, VLM, diffusion, transformers, RL
Cloud: AWS, Google Cloud
Programming: Python, C, C++, Git, Docker, Linux